

S&P Risk Premia Rates Indices *Methodology*

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Introduction

Index Objective and Highlights

S&P Risk Premia Rates Indices. The indices are weighted return indices that measure the performance of long-short risk premia strategies in seven major global 10-year government bond futures contracts. The markets include the U.S., Canada, UK, Germany, Japan, Australia, and France. The performance of each bond futures contract represents the performance of a quarterly rolling position in the front month futures contract, measured by each index in the S&P Global Bond Futures Index Series. All indices maintain long allocations in three futures contracts with the highest values and short allocations in three futures contracts with the lowest values.

S&P Risk Premia FX Indices. The indices measure the performance of long-short alternative risk premia strategies in G10 currencies. The performance of each currency represents the performance of a quarterly rolling investment in currency forward versus the U.S. dollar. The indices rebalance on International Monetary Market (“IMM”) dates, which are the third Wednesday of March, June, September, and December, respectively. The performance of a long position on each G10 currency forward is tracked by the relevant S&P G10 Currency Index. The relevant S&P G10 Currency Index stipulates that on each rebalancing date a currency forward contract is assumed to be entered with a settlement date that coincides with the spot settlement date on the next rebalancing date. On the next rebalancing date, this forward contract is rolled over into a new forward contract.

For information on the S&P Global Bond Futures Index Series, please refer to the S&P Futures Indices Methodology at www.spglobal.com/spdji/.

For information on the S&P G10 Currency Indices, please refer to Appendix A.

Index Family

The S&P Risk Premia Rates Indices family includes:

- **S&P Risk Premia Rates Carry Index.** The index measures the performance of a long-short carry strategy in major global government bond futures. Bond futures carry is measured by the yield spread between the 10-year yield and a short-term interest rate. The carry strategy maintains long allocations in the three futures contracts with the highest carry and short allocations in the three futures contracts with the lowest carry.
- **S&P Risk Premia Rates Value (Spread Reversion) Index.** The index measures the performance of a long-short value strategy in major global government bond futures. The value of bond futures is measured by the z-score of the Yield Spread between the 10-year yield and a short-term interest rate with respect to its one-year history. The value strategy maintains long allocations in the three futures contracts with the highest z-scores and short allocations in the three futures contracts with the lowest z-scores.
- **S&P Risk Premia Rates Momentum (Cross-Sectional) Index.** The index measures the performance of a long-short momentum strategy in major global government bond futures. Momentum is measured by the 12-month excess return index value change in local currency, excluding the most recent month. The momentum strategy maintains long allocations in the three futures contracts with the highest momentum and short allocations in the three futures with the lowest momentum.

The S&P Risk Premia FX Indices family includes:

- **S&P Risk Premia FX Value G10 Index.** The index measures the performance of a long-short value strategy consisting of G10 currencies. Currency value is measured by the ratio of spot foreign exchange rate and Organisation for Economic Co-operation and Development (OECD) purchasing power parities (PPP). The value strategy maintains long exposure in the three currencies with the lowest value ratios and short exposure in those three currencies with the highest value ratios. The indices are equally weighted within each strategy bucket.
- **S&P Risk Premia FX Carry G10 Index.** The index measures the performance of a long-short carry strategy consisting of G10 currencies. Currency carry is measured by each currency's three-month yield. The carry strategy maintains long exposure in the three currencies with the highest carry rankings and short exposure in the three currencies with the lowest carry rankings. Currencies that are long (short) are either assigned a weight of 50% or 30% or 20% (-50% or -30% or -20%) based on the ranking of the carry signal.

Supporting Documents

This methodology is meant to be read in conjunction with supporting documents providing greater detail with respect to the policies, procedures and calculations described herein. References throughout the methodology direct the reader to the relevant supporting document for further information on a specific topic. The list of the main supplemental documents for this methodology and the hyperlinks to those documents is as follows:

Supporting Document	URL
S&P Dow Jones Indices' Index Mathematics Methodology	Index Mathematics Methodology
S&P Dow Jones Indices' Commodity Index Mathematics Methodology	Commodity Index Mathematics Methodology
S&P Dow Jones Indices' Commodities Indices Policies & Practices Methodology	Commodities Indices Policies & Practices
S&P Dow Jones Indices' Index Governance Policies Methodology	Index Governance Policies Methodology

This methodology was created by S&P Dow Jones Indices to achieve the aforementioned objective of measuring the underlying interest of each index governed by this methodology document. Any changes to or deviations from this methodology are made in the sole judgment and discretion of S&P Dow Jones Indices so that the index continues to achieve its objective.

Index Construction

A long position in a bond futures contract is a measure of the performance of a long position in the underlying benchmark bond, earning an equivalent yield reduced by the funding rate required to fund the long position and adjusted by the cheapest to deliver option on the benchmark bond held by the counterparty. For simplicity we calculate the Yield Spread between 10-year government bond yield and a short-term interest rate to approximate the yield that government bond futures carry.

Calculate the *Yield Spread_i* for country *i* in the universe of 10-year government bond futures:

$$Yield\ Spread_i = 10\text{-year on-the-run government bond yield}_i - 3\text{-month interest rate}$$

Component Index	Underlying Futures Contract	Symbol	Exchange	Interest Rate
S&P 10-Year U.S. Treasury Note Futures	10-Year U.S. Treasury Note Futures	TY	CME	3M US Treasury Rate
S&P 10-Year Canada Government Bond Futures Index	10-Year CGB Futures	CGB	MX	3M Canada Treasury Rate
S&P Long Gilt Futures Index	Long Gilt Futures	FLG	ICE	3M UK Treasury Rate
S&P Euro-Bund Futures	Euro-Bund Futures	FGBL	EUREX	3M German Treasury Rate
S&P 10-Year JGB Futures	10-Year JGB Futures	JGB	JPX	3M Japan Treasury Rate
S&P/ASX Australian 10-Year Treasury Bond Futures	10-Year Australian Treasury Bond Futures	XT	ASX	Australian Dollar 3 Month Bank Bill
S&P Euro-OAT Futures	Euro-OAT Futures	FOAT	EUREX	3M France Treasury Rate

S&P Risk Premia Rates Carry Index

Weight $W_{i,rb}$ for the designated futures contract of country *i* effective after the close of the rebalancing date *rb* is calculated with the following steps:

1. Rank all the countries in the universe in ascending order by *Yield Spread_{i,rf}* as of the rebalancing reference dates *rf*
2. $W_{i,rb} = -1/3$ if rank of *Yield Spread_{i,rf}* ≤ 3
 $= +1/3$ if rank of *Yield Spread_{i,rf}* ≥ 5 ¹

S&P Risk Premia Rates Value (Spread Reversion) Index

Weight $W_{i,rb}$ for the designated futures contract of country *i* effective after the close of the rebalancing date *rb* is calculated with the following steps:

1. Calculate the z-score of *Yield Spread* as of the rebalancing reference dates *rf* as follows:

$$Z_{i,rf} = \frac{Yield\ Spread_{i,rf} - \mu_{i,rf}}{\sigma_{i,rf}}$$

where:

¹ The S&P Euro-OAT Futures Index has a first value date of 04/30/2012. For rebalancings prior to 05/31/2012, only six underlying indices were included in the backtest, and $W_{i,rb} = +1/3$ if rank ≥ 4 .

$\mu_{i,rf}$ = Arithmetic mean of the winsorized daily *Yield Spread* variable over the past 12 months as of *rf*

$\sigma_{i,rf}$ = Standard deviation of the winsorized daily *Yield Spread* variable over the past 12 months as of *rf*

Winsorization reduces the impact of outliers on a data set by limiting them to a designated value. Both $\mu_{i,rf}$ and $\sigma_{i,rf}$ above are calculated with 95% winsorization, i.e, all data over the last 12 months below the 2.5% percentile is set to the 2.5% percentile and data over the last 12 months above the 97.5% percentile is set to the 97.5% percentile.

2. Rank all the countries in the universe in ascending order by $Z_{i,rf}$

3. $W_{i,rb} = -1/3$ if rank of $Z_{i,rf} \leq 3$

$= +1/3$ if rank of $Z_{i,rf} \geq 5^2$

S&P Risk Premia Rates Momentum (Cross-Sectional) Index

Weight $W_{i,rb}$ for the designated futures contract of country *i* effective after the close of the rebalancing date *rb* is calculated with the following steps:

$$1. \text{Momentum}_{i,rf} = \left(\frac{ER_{i,prev,rf}}{ER_{i,prev,rf-12m}} \right) - 1$$

where:

$ER_{i,prev,rf}$ = excess return index value of the respective bond futures index *i* in local currency as of the previous month reference date

$ER_{i,prev,rf-12m}$ = excess return index value of the respective bond futures index *i* in local currency as of 12 months prior to the previous month reference date

2. Rank all the countries in the universe in ascending order by $\text{Momentum}_{i,rf}$

3. $W_{i,rb} = -1/3$ if rank of $\text{Momentum}_{i,rf}$ is ≤ 3

$= +1/3$ if rank of $\text{Momentum}_{i,rf}$ is $\geq 5^3$

S&P Risk Premia FX G10 Value Index

Currency Weight Calculation. Effective after closing of the rebalancing dates *rb*, currency weight $W_{i,rb}$ for currency *i* is calculated with the following steps:

1. Calculate the *Value_Ratio* for currency *i* for each currency in the G10 currency basket:

$$\text{ValueRatio}_{i,rb} = \frac{\text{Spot}_{i,rb}}{\text{PPP}_{i,rb}}$$

where:

$\text{Spot}_{i,rb}$ = The spot foreign exchange rate expressed in USD per currency as of the rebalancing date *rb*.

² The S&P Euro-OAT Futures Index's first value date is 04/30/2012. For rebalancings prior to 05/31/2013, only six underlying indices were included in the backtest, and $W_{i,rb} = +1/3$ if rank ≥ 4 .

³ The S&P Euro-OAT Futures Index's first value date is 04/30/2012. For rebalancings prior to 06/30/2013, only six underlying indices were included in the backtest, and $W_{i,rb} = +1/3$ if rank ≥ 4 .

$PPP_{i,rb}$ = The latest update of purchasing power parities (PPPs) for GDP published by the OECD each year in February for the previous year immediately preceding the current rebalancing. It is expressed in USD per currency.⁴

PPPs are the rates of currency conversion that equalize the purchasing power of different currencies by eliminating the differences in price levels between countries.⁵

By definition, this *Value_Ratio* is equal to one for the US\$ dollar.

2. Rank all G10 currencies (including US\$ dollar) by Value Ratio in ascending order.
3. $W_{i,rb} = 1/3$ if rank ≤ 3 , or $-1/3$ if rank ≥ 8

S&P Risk Premia FX G10 Carry Index

Currency Weight Calculation. Effective after closing of the rebalancing dates rb , currency weight $W_{i,rb}$ for currency i calculates with the following steps:

1. Determine the three-month (3M) yield for currency i for each currency in the G10 currency basket:
For the US Dollar, the 3M US T-Bill is used. For the remaining G10 currencies, the 3M yield is implied from the spot, FX forward exchange rate and 3M US T-Bill as follows:

$$Implied3MYield_{i,rf} = \left(\left(\left(\frac{Spot_{i,rf}}{Forward_{i,rf}} \right) * \left(1 + \frac{TBR_{i,rf}}{4} \right) \right) - 1 \right) * 4$$

where:⁶

$Spot_{i,rf}$ = The spot exchange rate expressed in USD per currency as of the reference date rf .

$Forward_{i,rf}$ = The 3M FX forward exchange rate expressed in USD per currency as of the reference date rf .

$TBR_{i,rf}$ = The 3M US T-Bill as of the reference date rf .

2. Rank all G10 currencies by 3M rate in descending order.
3. $W_{i,rb} = 0.5$ if rank = 1, $W_{i,rb} = 0.3$ if rank = 2 and $W_{i,rb} = 0.2$ if rank = 3
 $W_{i,rb} = -0.5$ if rank = 10, $W_{i,rb} = -0.3$ if rank = 9 and $W_{i,rb} = -0.2$ if rank = 8

In the case of a tie, apply the following rules:

- If 1 and 2 (10 or 9) are tied: 25% (-25%) weight each
- If 2 and 3 (9 or 8) are tied: 15% (-15%) weight each
- If 3 and 4 (8 or 7) are tied: 10% (-10%) weight each

⁴ PPPs are published by the OECD in terms of national currency per US dollar. PPPs are published and revised five times per annum at T+2 months, T+6 months, T+12 months, T+24 months, and T+35 months. Note that the historical back-test reflects all PPPs available as of 12/31/2018.

⁵ Source: <https://data.oecd.org/conversion/purchasing-power-parities-ppp.htm>.

⁶ 90/360 day convention is assumed for all the G10 currencies.

Excess Return Indices Calculation – Rates Indices

On each business date t , the excess return index (USD) is calculated as:

$$ERIV_t = ERIV_{rb} * \sum_i W_{i,rb} * \left(\frac{ER_{i,t}}{ER_{i,rb}} - 1 \right) * \left(\frac{FX_{i,t}}{FX_{i,rb}} \right)$$

where:

$ERIV_t$ = Excess Return Index value as of the current date t

$ERIV_{rb}$ = Excess Return Index value as of the last rebalancing date rb

$W_{i,rb}$ = Weight of futures contract i effective from the last rebalancing date rb

$ER_{i,t}$ = excess return index value of the respective bond futures index i in local currency as of t

$ER_{i,rb}$ = excess return index value of the respective bond futures index i in local currency as of the last rebalancing date rb

$FX_{i,t}$ = FX spot rate for the currency denomination of futures contract i as of t , in USD per currency

$FX_{i,rb}$ = FX spot rate for the currency denomination of futures contract i as of rb , in USD per currency

Both $ER_{i,t}$ and $ER_{i,rb}$ are the excess return index values from S&P Global Bond Futures Index Series

Excess Return Indices Calculation – FX Indices

On each business date t , the excess return index calculates as:

$$IndexER_t = IndexER_{rb} * \left(1 + \sum_i W_{i,rb} * \left(\frac{ER_{i,t}}{ER_{i,rb}} - 1 \right) \right)$$

where:

$IndexER_t$ = Excess return index value as of the current date t

$IndexER_{rb}$ = Excess return index value as of the last rebalancing date rb

$W_{i,rb}$ = Weight of currency i effective from the last rebalancing date rb

$ER_{i,t}$ = S&P G10 Currency Indices excess return index value as of t

$ER_{i,rb}$ = S&P G10 Currency Indices excess return index value as of the last rebalancing date rb

Total Return Indices Calculation – Rates Indices

On any given calendar day, t , the Treasury Bill Return (TBR) is equal to an amount determined in accordance with the following formula:

$$TBR_t = \left[\frac{1}{1 - \frac{91}{360} * TB_{t-1}} \right]^{\frac{1}{91}} - 1$$

where:

TB_{t-1} = The 3 month T-Bill Rate available on the business day prior to day t .

A total return (USD) version of each index is calculated, which includes interest accrual based on 3 month T-Bill rate, as follows:

$$TRIV_t = TRIV_{t-1} * (1 + EReturn_t + TBR_t) * (1 + TBR_t)^{days}$$

where:

$TRIV_t$ = Total return index value as of the current date, t .

$TRIV_{t-1}$ = Total return index value as of immediately preceding CME Business Day $t-1$.

$EReturn_t$ = Excess Return on date t from the immediately preceding CME Business Day $t-1$

$$= \frac{ERIV_t}{ERIV_{t-1}} - 1$$

TBR_t = the Treasury Bill Return on date t

days = Number of non CME business days since the prior CME Business Day $t-1$

For more information on the index calculation methodology, please refer to S&P Dow Jones Indices' Index Mathematics Methodology.

Total Return Indices Calculation – FX Indices

A total return version of each index, which includes interest accrual on the notional value of the index based on the overnight U.S. Fed Funds rate, calculates as follows:

$$IndexTR_t = IndexTR_{rb} * ER_t + IndexTR_{rb} * \prod_{rb+1}^t \frac{(1 + FFR_{t-1} * D_{t-1,t})}{360}$$

where:

$IndexTR_t$ = Total return index value as of the current date, t .

$IndexTR_{rb}$ = Total return index value as of the last rebalancing date, rb .

ER_t = Excess return from the prior rebalancing date to t .

$D_{t-1,t}$ = Count of calendar days from the previous trading date, $t-1$, to the current date, t .

FFR_{t-1} = Federal Reserve Overnight Bank Funding Rate as of the last trading date, $t-1$.

For more information on the index calculation methodology, please refer to the Other Derived Indices sections of the S&P Dow Jones Indices' Commodity Index Mathematics Methodology.

Index Maintenance

Rebalancing

S&P Risk Premia Rates Indices. The indices rebalance on month-end dates when all of the relevant futures markets are open for official trading. If any one of the relevant futures markets is not open, the indices rebalance on the immediately preceding day when all of the relevant futures markets are open for official trading. The rebalancing reference date is one business day, as determined by the CME holiday calendar, prior to the rebalancing date. All data required to rebalance the indices are obtained as of the rebalancing reference date. If any data on rebalancing reference date is not available, the latest data available prior to the rebalancing reference date is used.

Rebalancing Schedule	
Rebalancing Frequency	Monthly
Announcement Date	<i>T-1</i>
Rebalancing Reference Date	<i>T-1</i>

S&P Risk Premia FX Indices. The indices rebalance on International Monetary Market (“IMM”) dates, which are the third Wednesday of March, June, September, and December, respectively. (*T*) dates. The rebalancing reference date is two business day prior to the rebalancing date. All data required to rebalance the indices is obtained as of the rebalancing reference date.

Rebalancing Schedule	
Rebalancing Frequency	Quarterly
Rebalancing Dates	IMM dates
Announcement Date	<i>T - 1</i>
Rebalancing Reference Date	<i>T - 2</i>

All days reflect after the close and count business days.

Data Source

The source for on-the-run 10-year government bond yield rates is LSEG. The source data for FX spot rates is WMR London 4:00 PM fixing rates.

WMR foreign exchange rates are taken daily at 4:00 PM London Time and used in the calculation of the indices. These mid-market fixings are calculated by WMR based on LSEG data and appear on LSEG pages.

Index Governance

Index Committee

An S&P Dow Jones Indices' Index Committee maintains the indices. The Index Committee meets regularly. At each meeting, the Index Committee reviews any significant market events. In addition, the Index Committee may revise index policy covering rules for timing of rebalancing or other matters.

S&P Dow Jones Indices considers information about changes to its indices and related matters to be potentially market moving and material. Therefore, all Index Committee discussions are confidential.

S&P Dow Jones Indices' Index Committees reserve the right to make exceptions when applying the methodology if the need arises. In any scenario where the treatment differs from the general rules stated in this document or supplemental documents, clients will receive sufficient notice, whenever possible.

In addition to the daily governance of indices and maintenance of index methodologies, at least once within any 12-month period, the Index Committee reviews the methodology to ensure the indices continue to achieve the stated objectives, and that the data and methodology remain effective. In certain instances, S&P Dow Jones Indices may publish a consultation inviting comments from external parties.

For information on Quality Assurance and Internal Reviews of Methodology, please refer to S&P Dow Jones Indices' Equity Indices Policies & Practices Methodology and the S&P Dow Jones Indices' Index Governance Policies Methodology.

Index Policy

Holiday Schedule

The S&P Risk Premia Rate indices are calculated when CME is open. If any one of the relevant futures markets is not open, bond futures closing price for the previous day is used.

The S&P Risk Premia FX indices calculate when the International Money Market of the Chicago Mercantile Exchange (CME) is open.

Rebalancing

The Index Committee may change the date of a given rebalancing for reasons including market holidays occurring on or around the scheduled rebalancing date. Any such change will be announced with proper advance notice where possible.

Unexpected Exchange Closures

For information on Unexpected Exchange Closures, please refer to S&P Dow Jones Indices' Commodity Indices Policies & Practices Methodology.

Recalculation Policy

For information on the recalculation policy, please refer to S&P Dow Jones Indices' Commodity Indices Policies & Practices Methodology.

Contact Information

For any questions regarding an index, please contact: index_services@spglobal.com.

Index Dissemination

Index levels are available through S&P Dow Jones Indices' Web site at www.spglobal.com/spdji/, major quote vendors (see codes below), numerous investment-oriented Web sites, and various print and electronic media.

Tickers

The table below lists headline indices covered by this document. All versions of the below indices that may exist are also covered by this document. Please refer to the [S&P DJI Methodology & Regulatory Status Database](#) for a complete list of indices covered by this document.

Index Code	Index Name	Launch Date	Base Date	Base Value	BBG	RIC
SPRPRCP	S&P Risk Premia Rates Carry Index (USD) ER	02/10/2020	09/28/2001	100	SPRPRCP	SPRPRCP
SPRPRCT	S&P Risk Premia Rates Carry Index (USD) TR	02/10/2020	09/28/2001	100	SPRPRCT	SPRPRCT
SPRPRVP	S&P Risk Premia Rates Value (Spread Reversion) Index (USD)	02/10/2020	09/28/2001	100	SPRPRVP	SPRPRVP
SPRPRVT	S&P Risk Premia Rates Value (Spread Reversion) Index (USD)	02/10/2020	09/28/2001	100	SPRPRVT	SPRPRVT
SPRPRMUP	S&P Risk Premia Rates Momentum (Cross-Sectional) Index (USD) ER	02/10/2020	02/28/2002	100	SPRPRMUP	SPRPRMUP
SPRPRMUT	S&P Risk Premia Rates Momentum (Cross-Sectional) Index (USD) ER	02/10/2020	02/28/2002	100	SPRPRMUT	SPRPRMUT
SPRPFVGP	S&P Risk Premia FX Value G10 Index (USD) ER	02/10/2020	12/21/2005	100	SPRPFVGP	.SPRPFVGP
SPRPFVGT	S&P Risk Premia FX Value G10 Index (USD) TR	02/10/2020	12/21/2005	100	SPRPFVGT	.SPRPFVGT
SPRPF CGP	S&P Risk Premia FX Carry G10 Index (USD) ER	12/15/2020	12/21/2005	100	--	--
SPRPF CGT	S&P Risk Premia FX Carry G10 Index (USD) TR	12/15/2020	12/21/2005	100	--	--
SPAUD3MP	S&P AUD/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPAUD3MT	S&P AUD/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--
SPCAD3MP	S&P CAD/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPCAD3MT	S&P CAD/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--
SPCHF3MP	S&P CHF/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPCHF3MT	S&P CHF/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--
SPEUR3MP	S&P EUR/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPEUR3MT	S&P EUR/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--
SPGBP3MP	S&P GBP/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPGBP3MT	S&P GBP/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--
SPJPY3MP	S&P JPY/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPJPY3MT	S&P JPY/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--
SPNOK3MP	S&P NOK/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPNOK3MT	S&P NOK/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--
SPNZD3MP	S&P NZD/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPNZD3MT	S&P NZD/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--
SPSEK3MP	S&P SEK/USD 3M Forward Index ER	10/16/2019	12/21/2005	100	--	--
SPSEK3MT	S&P SEK/USD 3M Forward Index TR	10/16/2019	12/21/2005	100	--	--

Index Data

Daily index levels and data are available via subscription.

For product information, please contact S&P Dow Jones Indices, www.spglobal.com/spdji/en/contact-us.

Web site

For further information, please refer to S&P Dow Jones Indices' Web site at www.spglobal.com/spdji/.

Appendix A

S&P G10 Currency Indices Calculations

Traded Forward Rate Calculation

On each rebalancing date rb , the index assumes the existence of a currency forward contract with a settlement date coinciding with the spot settlement date on the next rebalancing date. This traded forward rate is obtained by a linear interpolation between the current three-month forward rate and the current spot rate.

$$TradedRate_{rb} = SpotRate_{rb} + (3MFwdRate_{rb} - SpotRate_{rb}) * \frac{Days\ in\ Trade}{Days\ in\ Period}$$

where:

$SpotRate_{rb}$ = The spot rate on rebalancing date t .

$Days\ in\ Trade$ = Calendar days between the spot settlement date on the next rebalancing date and the settlement date of the spot contracts on rebalancing date rb .

$Days\ in\ Period$ = Calendar days between the settlement date of the three-month forward contract on rebalancing date rb and the settlement date of the spot contract on rebalancing date rb .

Excess Return Index Calculation

On each rebalancing date rb , an ER index calculates by valuing the forward contract entered into on the prior rebalancing date using the current spot rate. On any other date t , the ER calculates by valuing the forward contract entered into on the prior rebalancing date using a linear interpolation between the current three-month forward rate and the current spot rate.

$$IndexER_t = IndexER_{rb} * (1 + ER_t)$$

where:

$IndexER_t$ = Excess return index value as of the current date t

$IndexER_{rb}$ = Excess return index value as of the last rebalancing date rb

ER_t = Excess return from the prior rebalancing date to t , calculated as follows:

$$ER_t = \frac{SpotRate_t}{Traded\ Rate_{rb}} - 1 \quad (\text{if } t \text{ is a rebalancing date})$$

$$ER_t = \frac{\left(\frac{Interpolated\ Fwd\ Rate_t}{Traded\ Rate_{rb}} - 1 \right)}{DF_t} \quad (\text{if } t \text{ is not a rebalancing date})$$

DF_t = Discount factor, which is defined with the following formula:

$$DF_t = \frac{(1 + (Discount\ Days))}{360 * SOFR_t}$$

$SOFR_t$ = SOFR Overnight + 014088% as of date t

$Traded\ Rate_{rb}$ = The traded rate on the last rebalancing date rb

$Interpolated\ Fwd\ Rate_t$ = The value, as of the close of date t , of the forward contract entered into the close of the prior rebalancing rb , calculates as follows:

$$Interpolated\ Fwd\ Rate_t = SpotRate_t + (3MFwdRate_t - SpotRate_t) * \frac{Days\ Left}{Days\ in\ Period}$$

If $Days\ Left > Days\ in\ Period$, $3MFwdRate_t$

$Spot\ Rate_t$ = Spot rate as of date t
 $3MFwdRate_t$ = Three-month forward rate as of date t
 $Days\ Left$ = Calendar days between the spot settlement date on the next rebalancing date and the settlement date of the spot contract on date t
 $Days\ in\ Period$ = Calendar days between the settlement date of the three-month forward contract on date t and the settlement date of the spot contract on date t

Except for USD/CAD, the G10 Currency Indices settle using the standard spot settlement on $T+2$. USD/CAD indices settle using the standard spot settlement on $T+1$.

Total Return Index Calculation

Each index calculates a TR version that includes interest accrual on the notional value of the index based on the overnight U.S. Fed Funds rate.

For information on TR index calculation, please refer to the Total Return Indices Calculation – FX Indices section above.

Rebalancing

The indices rebalance quarterly on IMM dates, the third Wednesday of March, June, September, and December, respectively.

All days reflect after the close and count business days.

Appendix B

Methodology Changes

Methodology changes since February 10, 2020, are as follows:

Change	Effective Date (After Close)	Methodology	
		Previous	Updated
Rates used for the calculation of rebalancing weights: 1. S&P 10-Year U.S. Treasury Note Futures 2. S&P 10-Year Canada Government Bond Futures Index 3. S&P Long Gilt Futures Index 4. S&P Euro-Bund Futures 5. S&P 10-Year JGB Futures 6. S&P Euro-OAT Futures	11/30/2021	1. US Dollar 3 Month ICE LIBOR 2. Canadian Dollar 3 Month Interest Rate Fixing CAD 3. UK Pound Sterling 3 Month ICE LIBOR 4. EURO 3 MONTH EURIBOR 5. Japanese Yen 3 Month ICE LIBOR 6. EURO 3 Month EURIBOR	1. 3M US Treasury Rate 2. 3M Canada Treasury Rate 3. 3M UK Treasury Rate 4. 3M German Treasury Rate 5. 3M Japan Treasury Rate 6. 3M France Treasury Rate

Disclaimer

Performance Disclosure/Back-Tested Data

Where applicable, S&P Dow Jones Indices and its index-related affiliates (“S&P DJI”) defines various dates to assist our clients by providing transparency. The First Value Date is the first day for which there is a calculated value (either live or back-tested) for a given index. The Base Date is the date at which the index is set to a fixed value for calculation purposes. The Launch Date designates the date when the values of an index are first considered live: index values provided for any date or time period prior to the index’s Launch Date are considered back-tested. S&P DJI defines the Launch Date as the date by which the values of an index are known to have been released to the public, for example via the company’s public website or its data feed to external parties. For Dow Jones-branded indices introduced prior to May 31, 2013, the Launch Date (which prior to May 31, 2013, was termed “Date of introduction”) is set at a date upon which no further changes were permitted to be made to the index methodology, but that may have been prior to the Index’s public release date.

Please refer to the methodology for the Index for more details about the index, including the manner in which it is rebalanced, the timing of such rebalancing, criteria for additions and deletions, as well as all index calculations.

Information presented prior to an index’s launch date is hypothetical back-tested performance, not actual performance, and is based on the index methodology in effect on the launch date. However, when creating back-tested history for periods of market anomalies or other periods that do not reflect the general current market environment, index methodology rules may be relaxed to capture a large enough universe of securities to simulate the target market the index is designed to measure or strategy the index is designed to capture. For example, market capitalization and liquidity thresholds may be reduced. In addition, forks have not been factored into the back-test data with respect to the S&P Cryptocurrency Indices. For the S&P Cryptocurrency Top 5 & 10 Equal Weight Indices, the custody element of the methodology was not considered; the back-test history is based on the index constituents that meet the custody element as of the Launch Date. Also, the treatment of corporate actions in back-tested performance may differ from treatment for live indices due to limitations in replicating index management decisions. Back-tested performance reflects application of an index methodology and selection of index constituents with the benefit of hindsight and knowledge of factors that may have positively affected its performance, cannot account for all financial risk that may affect results and may be considered to reflect survivor/look ahead bias. Actual returns may differ significantly from, and be lower than, back-tested returns. Past performance is not an indication or guarantee of future results.

Typically, when S&P DJI creates back-tested index data, S&P DJI uses actual historical constituent-level data (e.g., historical price, market capitalization, and corporate action data) in its calculations. As ESG investing is still in early stages of development, certain datapoints used to calculate certain ESG indices may not be available for the entire desired period of back-tested history. The same data availability issue could be true for other indices as well. In cases when actual data is not available for all relevant historical periods, S&P DJI may employ a process of using “Backward Data Assumption” (or pulling back) of ESG data for the calculation of back-tested historical performance. “Backward Data Assumption” is a process that applies the earliest actual live data point available for an index constituent company to all prior historical instances in the index performance. For example, Backward Data Assumption inherently assumes that companies currently not involved in a specific business activity (also known as “product involvement”) were never involved historically and similarly also assumes that companies currently involved in a specific business activity were involved historically too. The Backward Data Assumption allows the hypothetical back-test to be extended over more historical years than would be feasible using only actual data. For more information on “Backward Data Assumption” please refer to the FAQ. The methodology and factsheets of any index that employs backward assumption in the back-tested history will explicitly state so. The methodology will include an Appendix with a table setting forth the specific data points and relevant time period for which backward projected data was used. Index returns shown do not represent the results of actual trading of investable assets/securities. S&P DJI maintains the index and calculates the index levels and performance shown or discussed but does not manage any assets.

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